SHASHANK MOURYA

AI/ML Engineer · Quality Assurance & Automation

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SUMMARY

Information Technology undergraduate (CGPA 9.01) and aspiring AI Systems Engineer with live-project industry experience — two months at Nerumach.AI, progressing from QA / Automation Engineer to QA Lead. Builds production-grade systems across computer vision, machine learning and real-time WebXR, in C++17, Python and TypeScript. National hackathon winner (1st of 3,200+ teams) with a bias toward privacy-first architecture, measurable performance and testable engineering.

SKILLS

- **Languages:** C++ (C++17), Python, TypeScript / JavaScript, SQL
- **AI / ML:** Computer Vision, CNNs, XGBoost, Large Language Models (LLMs), Image Forensics
- **QA & Automation:** Manual & Functional Testing, Test Case Design, Regression & Smoke Testing, Automation Scripting, Bug Triage & Reporting, Agile / Sprint QA, Release Sign-off
- **Frameworks & Tools:** React, Three.js / React-Three-Fiber, WebXR, Vite, Git, REST APIs
- **Foundations:** Data Structures & Algorithms, Object-Oriented Programming, System Design
- **Soft Skills:** Team Leadership, Analytical Thinking, Technical Communication, Adaptability

EDUCATION

B.Tech, Information Technology — MMCOE, Pune (*2024 – 2028)*

CGPA: 9.01 / 10

EXPERIENCE

Nerumach.AI — QA / Automation Engineer → QA Lead (Live Project)*2 Months (16th June 2026 - 16th August 2026)*

Month 1 — QA / Automation Engineer

- Owned functional, regression and exploratory testing for a live production product, authoring structured test cases and reproducible defect reports across each sprint.
- Built and maintained automation scripts for repeatable regression suites, cutting manual verification effort per release cycle and catching regressions before they reached staging.
- Worked directly with developers to triage, reproduce and verify fixes, keeping the defect backlog closed out within the sprint.

Month 2 — QA Lead

- Promoted to QA Lead after one month; took ownership of the overall QA strategy, test plan and release quality for the project.
- Led and coordinated the QA effort — assigning test scope, reviewing peer test cases and consolidating reporting for stakeholders.

PROJECTS

LunarComm — WebXR Simulation of the ISRO–JAXA LUPEX Lunar Mission

React, TypeScript, Three.js/React-Three-Fiber, WebXR, Vite, Firebase Realtime Database, JavaScript

- **LunarComm** (Top 100 of 11,000+ teams): Developed a live WebXR-based Lunar South Pole simulation demonstrating RFC 9171 DTN protocol for rover communications in permanently shadowed regions (PSRs).
- Architected a Firebase-powered real-time state system implementing genuine store-and-forward bundle queuing to manage connectivity loss.
- Built a dual-interface demo with a mobile "rover" trigger and mission control dashboard featuring live connection visualization and redundancy.
- Created a browser-based WebXR environment for Meta Quest 2, rendering the lunar surface with physically-motivated lighting and interactive animations.
- Benchmarked the architecture against NASA LunaNet and SpaceX Starlink, proposing a robot-deployable ground relay mesh to solve "last-mile" line-of-sight gaps.

InsureTrust — Multi-Layered Forensic Insurance Verification

Python, XGBoost, Image Forensics, Machine Learning

- Developed an end-to-end forensic ecosystem that detects insurance fraud by analysing document integrity and image metadata.
- Engineered a multi-layered verification pipeline combining XGBoost risk scoring with image forensic algorithms to identify digital tampering.

Drishti AI — Computer Vision Navigation & Safety System

Computer Vision, Python, Multilingual NLP (Hinglish), Audio Processing

- Architected a real-time navigation assistance system using computer vision for obstacle detection and spatial awareness.
- Integrated a custom Hinglish audio guidance layer to deliver intuitive, multilingual safety alerts in diverse environments.
- Optimised detection models for low-latency inference, ensuring real-time reliability for safety-critical navigation.

FlowState — Privacy-First Productivity & Cognitive Load Predictor

Behavioral Telemetry, Machine Learning, User-Space Orchestration

- Built a user-space, privacy-first behavioural orchestrator that monitors cognitive load and predicts burnout risk from telemetry.
- Ran all machine learning locally so behavioural metrics never leave the user's machine.
- Dynamically adjusts the work environment in response to real-time cognitive stress indicators to maximise developer flow.

Impulse Route — National-Level AI Routing Solution

AI Algorithms, Data Structures, Path Optimisation

- Engineered a high-efficiency routing engine for complex logistics, focused on path optimisation and resource allocation.
- Refined algorithmic performance for large-scale datasets, ensuring scalability for national-level deployment.
- Secured a place in the Top 77 of 1,600+ teams at the IGNISIA '26 National-Level AI Hackathon.

ACHIEVEMENTS & AWARDS

- **Winner — MIT ADT AI Grand Challenge 2026:** 1st place among 3,200+ teams nationwide for a high-impact AI prototype; awarded a ₹1,00,000 cash prize for technical excellence and innovation in AI systems.
- **National Finalist — IGNISIA '26 AI Hackathon:** Top 77 of 1,600+ teams (top 5%) across India for the Impulse Route project.
- **Top 6 Finalist — XENIA Hackathon 2026:** Recognised among 350+ competing teams for engineering excellence in rapid prototyping.
- **Top 100 Finalist — FarAway 2026:** Recognized among 11000+ competing teams for (top 0.9%) across India for engineering excellence.

LEADERSHIP & VOLUNTEERING

IT Tech Club, MMCOE — AI/ML Head *(August 2026 - Present)*

- Selected to head the AI/ML vertical of the college IT Tech Club, owning the technical direction of the club's AI initiatives.
- Plans and runs hands-on sessions, workshops and project mentorship in machine learning and computer vision for club members.
- Mentors juniors through project builds and hackathon preparation, growing an active AI/ML contributor base within the club.

MicDrop Club — Event Management Head (*Apr 2025 – Sep 2025*)

- **Strategic Planning:** Managed all logistics and operations for club events end to end, from setup through wrap-up.
- **Team Leadership:** Led a cross-functional team covering technical needs, equipment coordination and scheduling, improving live-shoot workflow.
- **Crisis Management:** Served as primary escalation point during fast-paced events, holding production quality and timelines.

ICSFT 2026 — Student Volunteer (*24 – 25 Apr 2026*)

- Supported the International Conference on Sustainable and Futuristic Technologies, technically sponsored by the IEEE Pune Section.
- Assisted in executing technical sessions on "AI Integration for Sustainable Development," engaging with global research trends.
- Collaborated with faculty convenors and technical partners including Vintech and EduLife Software to ensure seamless operations.

UDAAN 2K25 — Student Coordinator (*22 Mar 2025*)

- Owned scoring and final results, ensuring 100% accuracy in data handling for hundreds of participants.
- Automated data entry with Advanced Excel, reducing processing time by roughly two hours.